

Lecture No (5)

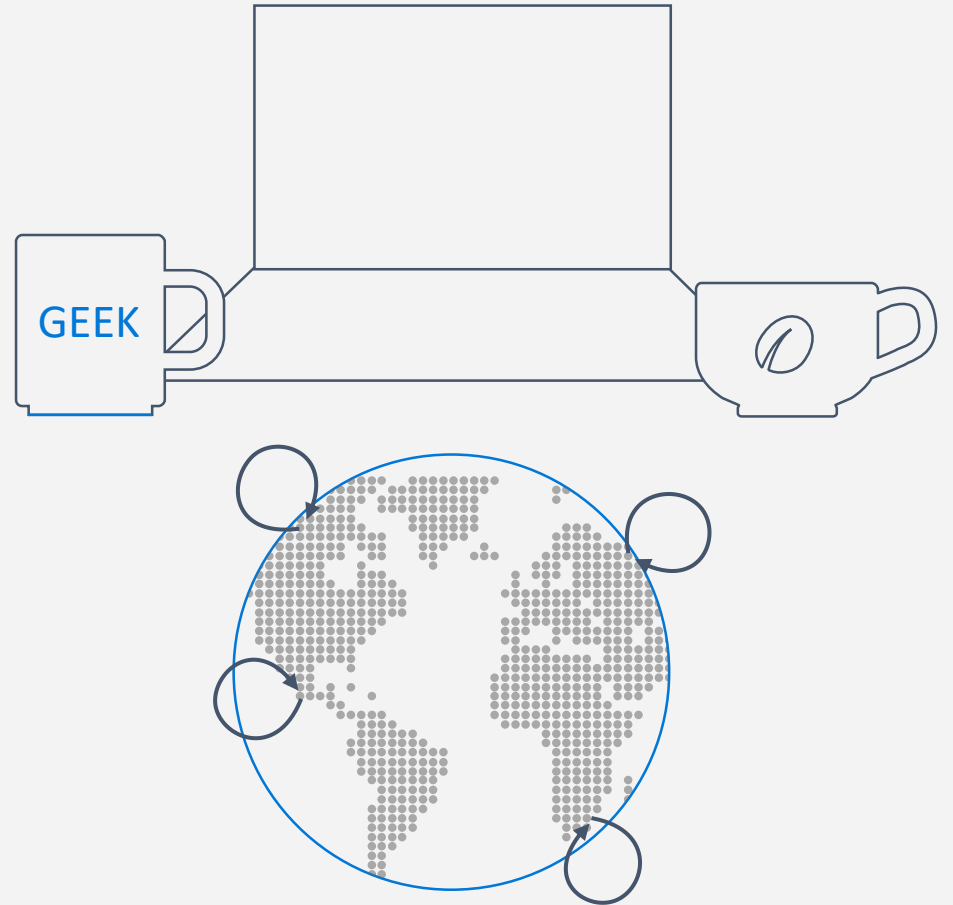
Server-Side Programming



Programming

Run native JavaScript server-side programming logic to perform atomic multi-record transactions.

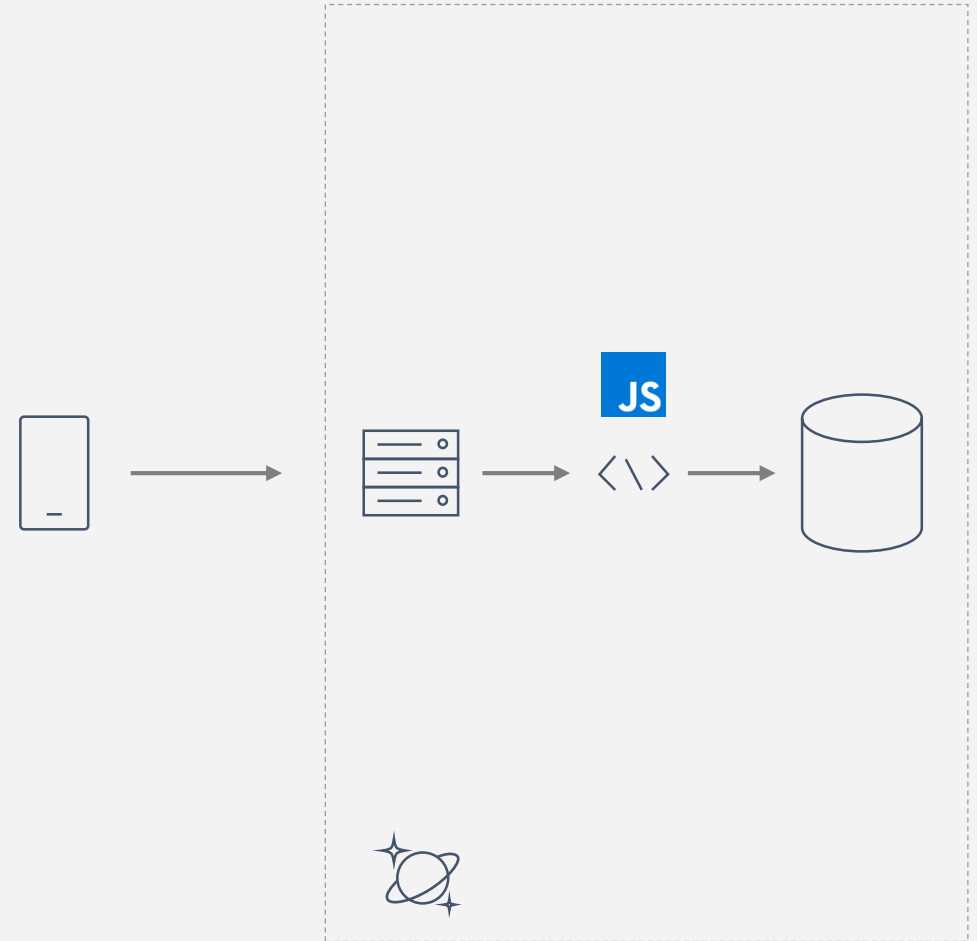
This module will reference programming in the context of the SQL API.



Stored Procedures

Benefits

- Familiar programming language
- Atomic Transactions
- Built-in Optimizations
- Business Logic Encapsulation

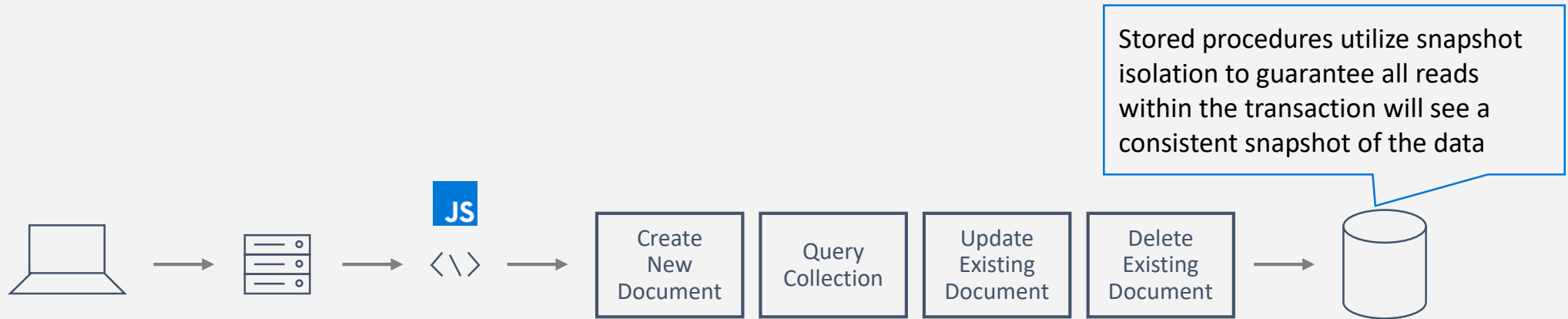


Multi-Document Transactions

Database Transactions

In a typical database, a transaction can be defined as a sequence of operations performed as a single logical unit of work. Each transaction provides ACID guarantees.

In Azure Cosmos DB, JavaScript is hosted in the same memory space as the database. Hence, requests made within stored procedures and triggers execute in the same scope of a database session.



Bounded Execution

Execution Within Time Boundaries

All Azure Cosmos DB operations must complete within the server-specified request timeout duration. If an operation does not complete within that time limit, the transaction is rolled back.

Helper Boolean Value

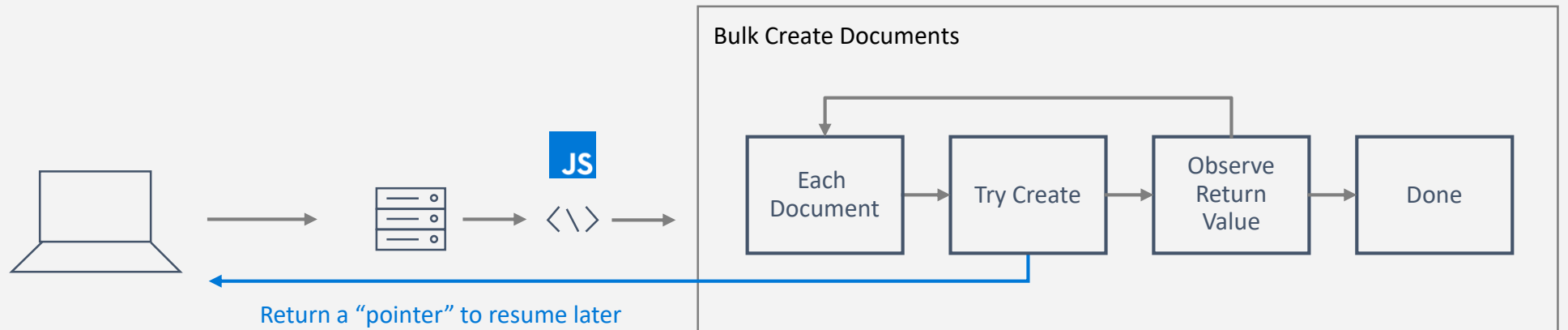
All functions under the collection object (for create, read, replace, and delete of documents and attachments) return a **Boolean value** that represents whether that operation will complete:

- **If true**, the operation is expected to complete
- **If false**, the time limit will soon be reached and your function should end execution as soon as possible.

Transaction Continuation Model

CONTINUING LONG-RUNNING TRANSACTIONS

- JavaScript functions can implement a continuation-based model to batch/resume execution
- The continuation value can be any value of your own choosing. This value can then be used by your applications to resume a transaction from a new “starting point”



CONTROL FLOW

Java script Control Flow

Stored procedures allow you to naturally express control flow, variable scoping, assignment, and integration of exception handling primitives with database transactions directly in terms of the JavaScript programming language.

ES6 Promises

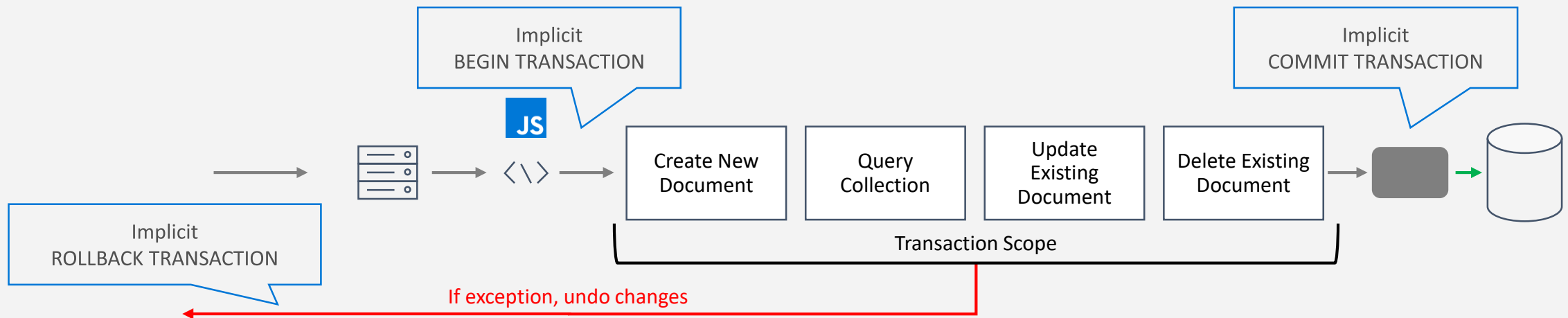
ES6 promises can be used to implement promises for Azure Cosmos DB stored procedures. Unfortunately, **promises “swallow” exceptions by default**. It is recommended to use callbacks instead of ES6 promises.

Rolling Back Transactions

Transaction Roll-Back

Inside a JavaScript function, all operations are automatically wrapped under a single transaction:

- If the function completes without any exception, all data changes are committed
- If there is any exception that's thrown from the script, Azure Cosmos DB's JavaScript runtime will roll back the whole transaction.



Debugging Stored Procedures

CONSOLE LOGGING

Much like with traditional JavaScript applications, you can use **console.log()** to capture various telemetry and data points for your running code.

VIEWING SCRIPT LOGS

.NET

You must opt-in to viewing and capturing console output using the Enable Script Logging boolean property available in the client SDK. The SDK has a ScriptLog property on the **Stored Procedure Response** class that contains the captured output of the JavaScript console log.

The End